Lecture 11 - The Chain Rule

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Quiz Time

Let’s get into neat rows, spread out, and slay this quiz! :)

The Chain Rule

The Road So Far

Recall we're trying to build a *synthetic* approach to studying derivatives:
Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

What is the Chain Rule? I

We're spending today building up the chain rule, the tool that will help us compute derivatives of the form:

$$\frac{d}{dx}(g(f(x))) \tag{1}$$

But what kinds of functions will this actually help us compute derivative of?
Sum/Difference Rule and Product/Quotient rule have visible signifiers that help us determine when to use them:

But what do functions of the form $g(f(x))$ even look like?

What is the Chain Rule? II

Recall our “basic” functions:

- x^n
- c^x
- $\sin(x), \cos(x)$, etc
- $|x|$
- $\log_c(x)$

When we *compose* functions (do the $g(f(x))$ thing), what does this actually do?
We make one function’s output another function’s input instead of just inputting x!

What is the Chain Rule? III

So, before we even *use* chain rule, we should first come up with a strategy for:

1. Recognizing when functions are composite functions
2. Recognizing which function is the inner function (the $f(x)$ in the $g(f(x))$) and which function is the outer function (the $g(x)$ in the $g(f(x))$)

Strategy:

1. Break up the complicated function into the addition, subtraction, product, and quotient of simpler functions (i.e. do sum/difference and product/quotient rules first)
2. In each of the smaller pieces, identify where the variables actually are.
3. At each occurrence of a variable, find the biggest/longest expression that it lives in - this is the innermost function.
4. Treat that whole expression as a new variable, and repeat from step 1.

Identifying Composites Example

For instance, let's take $h(t) = 3\sin^2(4t^2 + 1) \cdot \cos(4t^2 + 1) \cdot 8t$

Deriving Chain Rule I

Now let's assume we've already identified some function of interest as a composite $g(f(x))$, we'd like to find an expression of the derivative $\frac{d}{dx}g(f(x))$ hopefully in terms of derivatives involving $f(x)$ and $g(x)$.
Strategy: Linearize once again! Let's linearize around some $x = x_0$.
We start by linearizing *only* $f(x)$, since our input to $g(x)$ is changed by f and thus might end up close to a point different than $x = x_0$.
We know that $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$, so we end up wanting to further linearize:

$$g(f(x_0) + f'(x_0)(x - x_0))$$

Deriving Chain Rule II

Note: when we linearize $f(x)$ near $x = x_0$, we mean that for values $x \approx x_0$ we'll know that $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$ i.e. our input value being close to something implies our output value is close to the related thing.
For

$$g(f(x_0) + f'(x_0)(x - x_0))$$

what value are we inputting into the function g?
When $x \approx x_0$, what is $f(x_0) + f'(x_0)(x - x_0)$ close to?
Since $f(x_0) + f'(x_0)(x - x_0) \approx f(x_0)$, we will linearize $g(x)$ around $f(x_0)$ instead!

Deriving Chain Rule III

$$
\begin{aligned}
g(f(x)) &\approx g(f(x_0) + f'(x_0)(x - x_0)) \quad \text{linearize } g \text{ around } f(x_0) && (2)\\
&\approx g(f(x_0)) + g'(f(x_0))((f(x_0) + f'(x_0)(x - x_0)) - f(x_0)) && (3)\\
&= g(f(x_0)) + g'(f(x_0))(f'(x_0)(x - x_0) + f(x_0) - f(x_0)) && (4)\\
&= g(f(x_0)) + g'(f(x_0))f'(x_0)(x - x_0) && (5)
\end{aligned}
$$

Deriving Chain Rule IV

The derivative is the slope of the best linear approximation $\implies$

The Chain Rule

$$\frac{d}{dx}g(f(x)) = g'(f(x))f'(x) \tag{6}$$

Or, written differently

The Chain Rule (Leibniz's Version)

$$\frac{d(g \circ f)}{dx}\Big|_{x=x_0} = \frac{dg}{dx}\Big|_{x=f(x_0)} \cdot \frac{df}{dx}\Big|_{x=x_0} \tag{7}$$

Chain Rule Example I

Let's compute the derivative of that function from before:

$$h(t) = 3\sin^2(4t^2 + 1) \cdot \cos(4t + 1) \cdot 8t \tag{8}$$

Chain Rule Example II

We can also derive the quotient rule in terms of the chain rule since

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} = f(x) \cdot (\frac{1}{x} \circ g)(x) \tag{9}$$

i.e. $\frac{1}{g(x)}$ is a composite function! You first do $g(x)$ and then do the function $\frac{1}{x}$.
Then:

$$\frac{d}{dx}\frac{f(x)}{g(x)} = \left(\frac{d}{dx}f(x)\right) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{d}{dx}\frac{1}{g(x)}\right) \tag{10}$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \left(\left(\frac{d}{du}\frac{1}{u}\right)|_{u=g(x)} \cdot \left(\frac{d}{dx}g(x)\right)\right) \tag{11}$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \frac{-1}{g(x)^2} \cdot g'(x) = \frac{f'(x)g(x)}{g(x)^2)} - \frac{f(x)g'(x)}{g(x)^2} \tag{12}$$

$$= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{13}$$

Chain Rule Example III

We can also frame this more in terms of real world applications, too long for a slide but it's in the worksheet.

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!